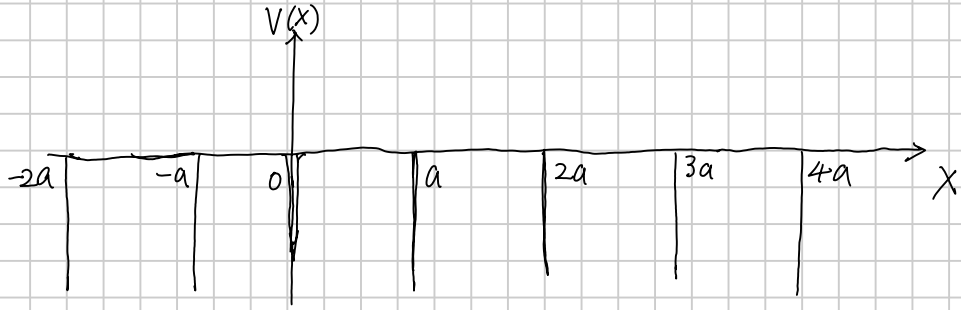
Solutions to HW11

Note Title 4/20/2009

5.20 $$V(x) = \alpha \sum_{j=0}^{N-1} \delta(x-ja), \qquad \alpha<0,\ j\in\mathbb{Z}$$




Schrodinger eq.

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi = E\psi$$

At $0<x<a$, $V(x)=0$

$$\Rightarrow \quad \frac{d^2\psi}{dx^2} = -k^2\psi \qquad \text{where } k \equiv \sqrt{\frac{2mE}{\hbar^2}}$$ , E can be positive or negative.

Let $\psi = A\sin kx + B\cos kx \qquad (0<x<a)$

According to Bloch's Theorem.

$$\psi(x+a) = e^{ika}\psi(x) \qquad (0<x<a)$$

$$\Rightarrow \quad \psi(y) = e^{ika}\left[A\sin k(y-a) + B\cos k(y-a)\right] \qquad (a<y<2a)$$

1° $\psi(x)$ is continuous @ $x=a$ $\Rightarrow$

$$A\sin ka + B\cos ka = e^{ika}\left[A\cdot 0 + B\right]$$

$$\Rightarrow \quad A = \frac{e^{ika} - \cos ka}{\sin ka} B \qquad (1)$$

2° $$-\frac{\hbar^2}{2m}\left[\psi(a+) - \psi(a-)\right] + \alpha\psi(a) = 0$$

$$\Leftrightarrow \quad -\frac{\hbar^2}{2m}\left[e^{ika}\cdot kA - kA\cos ka + kB\sin ka\right] + \alpha B e^{ika} = 0 \qquad (2)$$

substitute eq(1) into eq(2), we get (after simplification)

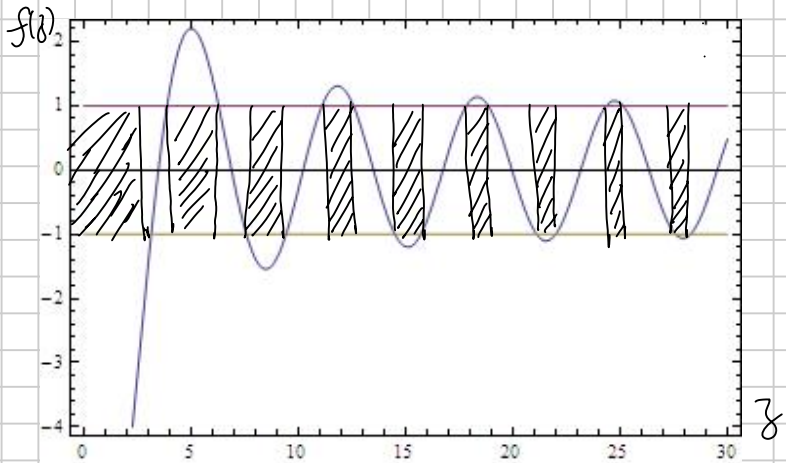
$$\cos(ka) = \cos(ka) + \frac{m\alpha}{\hbar^2 k}\sin(ka) \qquad (3)$$

a) If $E>0$, $k$ is real

Let $z \equiv ka$ & $\beta \equiv \frac{m\alpha a}{\hbar^2}$, the RHS of the above eq. can be written as

$$f(z) = \cos(z) + \beta \frac{\sin(z)}{z} \quad (4)$$

since $\alpha < 0$, $\Rightarrow \beta < 0$. $|\beta|$ is a dimensionless measure of the strength of the delta function. For the case of $\beta = -10$,




The argument will be the same with the argument in the text-book.
In the below, we will focus on negative energy.

b). If $E < 0$, $k$ is imaginary, $k = \pm i\sqrt{\frac{-2mE}{\hbar^2}} = \pm ip$.

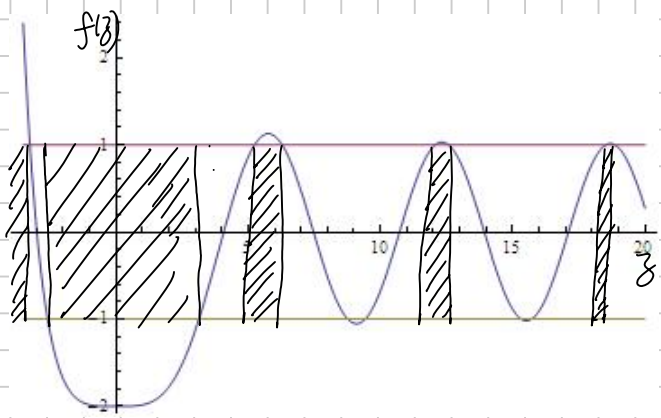
$$\cos(ka) = \cos(\pm ipa) = \cosh(pa)$$
$$\sin(ka) = \sin(\pm ipa) = \pm i\sinh(pa)$$

Now, the eq③ can be rewritten as.

$$\cos ka = \cosh(pa) + \frac{m\alpha}{\hbar^2 k}\sinh(pa)$$

let $\beta \equiv \frac{m\alpha}{\hbar^2 k} < 0$, $z = -pa$

$$f(z) \equiv \cosh(+z) + \beta \frac{\sinh(z)}{z} \quad (5)$$




Here $\beta = -3$

$z > 0$ for eq ④

$z < 0$ for eq ⑤

allowed energies are allowed in the range $-1 \le f(z) \le 1$

since $\cos(ka) = \cos\left(\frac{2\pi n}{N} a\right)$, $n = 0, 1, \cdots N-1$.

Obviously, the $1^{st}$ band contains $N$ states, so do all higher bands.

7-6. a) Assume the dwarf is a uniform solid sphere w/ Radius $R$.

Self energy $E = -\int \frac{G \cdot \rho \cdot \frac{4}{3}\pi r^3}{r} \cdot (\rho 4\pi r^2) dr$

$$= -G\rho^2 \frac{16\pi^2}{3} \int r^4 dr$$

$$= -G\rho^2 \frac{16}{3}\pi^2 \cdot \frac{1}{5} R^5$$

$$= -\frac{GM^2}{R} \cdot \frac{3}{5}$$

b). $\frac{2V}{(2\pi\hbar)^3} \frac{4}{3}\pi p_F^3 = N$ where $V = \frac{4}{3}\pi R^3$

$$\Rightarrow \quad P_F = \left[\frac{(2\pi\hbar)^3 \cdot N}{2 \cdot \frac{16}{9}\pi^2 R^3}\right]^{\frac{1}{3}} = \frac{9\pi}{4} N^{\frac{1}{3}} \frac{\hbar}{R}$$

$$E_K \sim N\frac{P_F^2}{2m} = \frac{81\pi^2}{16} \frac{N^{\frac{5}{3}}\hbar^2}{2mR^2} \sim \frac{\hbar^2 N^{\frac{5}{3}}}{mR^2} \approx \frac{\hbar^2 M^{5/3}}{m M_H^{5/3} R^2}$$

$$N \cdot M_H = M \Rightarrow N = \frac{M}{M_H}$$

↳ the mass of electron $\ll M_H$.

c) $$\frac{GM^2}{R} \approx \frac{\hbar^2 M^{5/3}}{m M_H^{5/3} R^2}$$

$$\Rightarrow \quad G M^{\frac{1}{3}} R \approx \frac{\hbar^2}{m M_H^{5/3}} \quad \text{(A)}$$

$$M^{\frac{1}{3}} R \approx \frac{\hbar^2}{G m M_H^{5/3}} \approx 10^{20}\, g^{1/3}\, cm.$$

d). $M \approx 2 \times 10^{33}\, g \Rightarrow R \approx 10^9\, cm$

$$\rho = \frac{M}{\frac{4}{3}\pi R^3} \approx \frac{10^{33}\, g}{10^{27}\, cm^3} \approx 10^6\, g/cm^3 \quad \text{(B)}$$

e). For neutron gas, eq (A) should be modified to

$$M^{\frac{1}{3}} R \approx \frac{\hbar^2}{G M_n M_H^{5/3}} = \frac{\hbar^2}{G M M_H^{5/3}} \cdot \frac{m}{M_n} \approx 10^{20} \times 10^{-3}\, g^{1/3} cm$$
$$= 10^{17}\, g^{1/3} cm$$

where $M_n$ is the mass of neutron, which is about 2000 times bigger the the mass of electron.

From eq (B) we see $R \approx 10^8$ cm $= 10$ km

$\rho \approx 10^{15}$ g/cm$^3$

7.8

$$E = \frac{V}{(2\pi\hbar)^3}\int 4\pi p^2 dp \cdot f_p(\epsilon)\cdot\epsilon$$

$$= \frac{V}{(2\pi\hbar)^3}\int dp \cdot 4\pi p^2 \cdot \frac{\epsilon}{e^{\frac{\epsilon(p)-\mu}{\tau}}-1}$$

$$\epsilon(p) = \frac{p^2}{2m}$$

$$d\epsilon = \frac{p}{m}dp$$

$$4\pi p^2 dp = 4\pi p m \underbrace{\frac{p}{m}dp} = 4\pi m\cdot(2m\epsilon)^{\frac{1}{2}} d\epsilon$$

$$E = \frac{V}{(2\pi\hbar)^3}\cdot 4\pi m\cdot(2m)^{\frac{1}{2}}\int d\epsilon \frac{\epsilon^{\frac{3}{2}}}{e^{\frac{\epsilon-\mu}{\tau}}-1}$$

let $\epsilon = x\tau$

$$E(\tau) = \frac{V}{4\pi^2}\left(\frac{2M}{\hbar^2}\right)^{3/2}\tau^{5/2}\int dx \frac{x^{3/2}}{\lambda e^x - 1}, \quad \text{where } \lambda = e^{-\frac{\mu}{\tau}}$$

where $\mu$ can be determined by the equation of total number $= N$.

heat capacity

$$C_V(\tau) = \left(\frac{\partial E(\tau)}{\partial \tau}\right)_{V,N}$$

and the entropy $\sigma$ via the eq: $C_V = \tau(\partial\sigma/\partial\tau)$